A learning image keeps the input origins of an internal response

Real input → selected ViT response → contributions at the original input locations



Input image

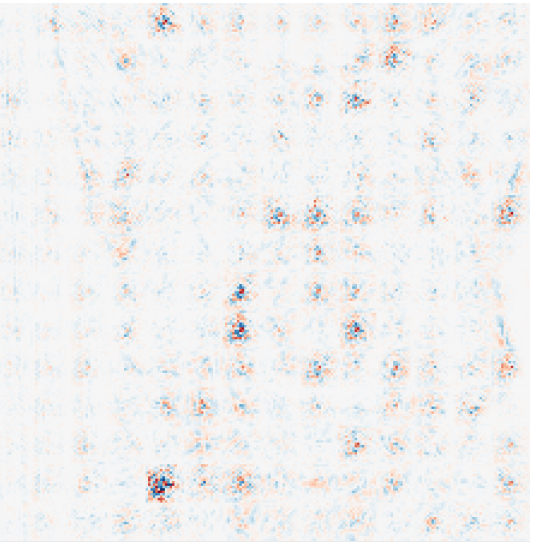
Trained ViT · Block 0

Patch embedding → Attention

Channel 0 · 196 patch tokens

$$s(x) = \sum_{t=1}^{196} O_{t,0}(x)$$

Response change Δs = -103.4506



One learning image · 224 × 224

− +

Signed path contribution · white = zero

$$L_s(i) = (x_i - x_{0i}) \int_0^1 \partial_i s(x_0 + \alpha(x - x_0))\, \mathrm{d}\alpha$$

Sum of raw map = -103.4506

One fixed input and channel · gray reference · full attention derivatives · no spatial smoothing